

we compute the cohomology with $\mathbb{Z}_p$ coefficients for p an odd prime. The answer will again be a polynomial algebra, but this time on even-dimensional generators, depending on the parity of n . Consider first the case that n is odd, say $n = 2k + 1$. There are two fiber bundles

$$V_2(\mathbb{R}^{2k+1}) \rightarrow E \rightarrow \tilde{G}_{2k+1}(\mathbb{R}^\infty) \quad V_2(\mathbb{R}^\infty) \rightarrow E \rightarrow \tilde{G}_{2k-1}(\mathbb{R}^\infty)$$

where E is the space of triples (P, v_1, v_2) with P an oriented $(2k + 1)$ -plane in $\mathbb{R}^\infty$ and v_1 and v_2 two orthogonal unit vectors in P . The projection map in the first bundle is $(P, v_1, v_2) \mapsto P$, and for the second bundle the projection sends (P, v_1, v_2) to the oriented $(2k - 1)$ -plane in P orthogonal to v_1 and v_2 , with the orientation specified by saying for example that v_1, v_2 followed by a positively oriented basis for the orthogonal $(2k - 1)$ -plane is a positively oriented basis for P . Both bundles are orientable since their base spaces $\tilde{G}_n(\mathbb{R}^\infty)$ are simply-connected, from the bundle $SO(n) \rightarrow V_n(\mathbb{R}^\infty) \rightarrow \tilde{G}_n(\mathbb{R}^\infty)$.

The fiber $V_2(\mathbb{R}^\infty)$ of the second bundle is contractible, so E has the same cohomology as $\tilde{G}_{2k-1}(\mathbb{R}^\infty)$. The fiber of the first bundle has the same $\mathbb{Z}_p$ cohomology as S^{4k-1} if p is odd, by the calculation at the end of §3.D. So if we assume inductively that $H^*(\tilde{G}_{2k-1}(\mathbb{R}^\infty); \mathbb{Z}_p) \approx \mathbb{Z}_p[p_1, \dots, p_{k-1}]$ with $|p_i| = 4i$, then Proposition 4D.11 above implies that $H^*(\tilde{G}_{2k+1}(\mathbb{R}^\infty); \mathbb{Z}_p) \approx \mathbb{Z}_p[p_1, \dots, p_k]$ where $p_k = e$ has dimension $4k$. The induction can start with $\tilde{G}_1(\mathbb{R}^\infty)$ which is just S^∞ since an oriented line in $\mathbb{R}^\infty$ contains a unique unit vector in the positive direction.

To handle the case of $\tilde{G}_n(\mathbb{R}^\infty)$ with $n = 2k$ even, we proceed just as in Example 4D.12, considering the bundles

$$S^{2k-1} \rightarrow E \rightarrow \tilde{G}_{2k}(\mathbb{R}^\infty) \quad S^\infty \rightarrow E \rightarrow \tilde{G}_{2k-1}(\mathbb{R}^\infty)$$

By the case n odd we have $H^*(\tilde{G}_{2k-1}(\mathbb{R}^\infty); \mathbb{Z}_p) \approx \mathbb{Z}_p[p_1, \dots, p_{k-1}]$ with $|p_i| = 4i$, so the corollary implies that $H^*(\tilde{G}_{2k}(\mathbb{R}^\infty); \mathbb{Z}_p)$ is a polynomial ring on these generators and also a generator in dimension $2k$.

Summarizing, for p an odd prime we have shown:

$$\begin{aligned} H^*(\tilde{G}_{2k+1}(\mathbb{R}^\infty); \mathbb{Z}_p) &\approx \mathbb{Z}_p[p_1, \dots, p_k], \quad |p_i| = 4i \\ H^*(\tilde{G}_{2k}(\mathbb{R}^\infty); \mathbb{Z}_p) &\approx \mathbb{Z}_p[p_1, \dots, p_{k-1}, e], \quad |p_i| = 4i, \quad |e| = 2k \end{aligned}$$

The same result holds also with $\mathbb{Q}$ coefficients. In fact, our proof applies for any coefficient ring in which 2 has a multiplicative inverse, since all that is needed is that $H^*(V_2(\mathbb{R}^{2k+1}); R) \approx H^*(S^{4k-1}; R)$. For a calculation of the cohomology of $\tilde{G}_n(\mathbb{R}^\infty)$ with $\mathbb{Z}$ coefficients, see [VBKT]. It turns out that all torsion elements have order 2, and modulo this torsion the integral cohomology is again a polynomial ring on the generators p_i and e . Similar results hold also for the cohomology of the unoriented Grassmann manifold $G_n(\mathbb{R}^\infty)$, but with the generator e replaced by p_k when $n = 2k$.

Exercises

1. By Exercise 35 in §4.2 there is a bundle $S^2 \rightarrow \mathbb{CP}^3 \rightarrow S^4$. Let $S^2 \rightarrow E_k \rightarrow S^4$ be the pullback of this bundle via a degree k map $S^4 \rightarrow S^4$, $k > 1$. Use the Leray-Hirsch theorem to show that $H^*(E_k; \mathbb{Z})$ is additively isomorphic to $H^*(\mathbb{CP}^3; \mathbb{Z})$ but has a different cup product structure in which the square of a generator of $H^2(E_k; \mathbb{Z})$ is k times a generator of $H^4(E_k; \mathbb{Z})$.
2. Apply the Leray-Hirsch theorem to the bundle $S^1 \rightarrow S^\infty/\mathbb{Z}_p \rightarrow \mathbb{CP}^\infty$ to compute $H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ from $H^*(\mathbb{CP}^\infty; \mathbb{Z}_p)$.
3. Use the Leray-Hirsch theorem as in Corollary 4D.3 to compute $H^*(V_n(\mathbb{C}^k); \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[x_{2k-2n+1}, x_{2k-2n+3}, \dots, x_{2k-1}]$ and similarly in the quaternionic case.
4. For the flag space $F_n(\mathbb{C}^n)$ show that $H^*(F_n(\mathbb{C}^n); \mathbb{Z}) \approx \mathbb{Z}[x_1, \dots, x_n]/(\sigma_1, \dots, \sigma_n)$ where σ_i is the i^{th} elementary symmetric polynomial.
5. Use the Gysin sequence to show that for a fiber bundle $S^k \rightarrow S^m \xrightarrow{p} S^n$ we must have $k = n - 1$ and $m = 2n - 1$. Then use the Thom isomorphism to show that the Hopf invariant of p must be ± 1 . [Hence $n = 1, 2, 4, 8$ by Adams' theorem.]
6. Show that if M is a manifold of dimension $2n$ for which there exists a fiber bundle $S^1 \rightarrow S^{2n+1} \rightarrow M$, then M is simply-connected and $H^*(M; \mathbb{Z}) \approx H^*(\mathbb{CP}^n; \mathbb{Z})$ as rings. Conversely, if M is simply-connected and $H^*(M; \mathbb{Z}) \approx H^*(\mathbb{CP}^n; \mathbb{Z})$ as rings, show there is a bundle $S^1 \rightarrow E \rightarrow M$ where $E \simeq S^{2n+1}$. [When $n > 1$ there are examples where M is not homeomorphic to $\mathbb{CP}^n$.]
7. Show that if a disk bundle $D^n \rightarrow E \rightarrow B$ has a Thom class with $\mathbb{Z}$ coefficients, then it is orientable.
8. If E is the product bundle $B \times D^n$ with B a CW complex, show that the Thom space $T(E)$ is homotopy equivalent to the n -fold reduced suspension $\Sigma^n B$, and that the Thom isomorphism specializes to the suspension isomorphism $H^i(B; R) \approx \tilde{H}^{n+i}(\Sigma^n B; R)$ given by the reduced cross product in §3.2.
9. Show that the inclusion $T^n \hookrightarrow U(n)$ of the n -torus of diagonal matrices is homotopic to the map $T^n \rightarrow U(1) \hookrightarrow U(n)$ sending an n -tuple of unit complex numbers $(z_1, \dots, z_n)$ to the 1×1 matrix $(z_1 \cdots z_n)$. Do the same for the diagonal subgroup of $Sp(n)$. [Hint: Diagonal matrices in $U(n)$ are compositions of scalar multiplication in n lines in $\mathbb{C}^n$, and $\mathbb{CP}^{n-1}$ is connected.]
10. Fill in the details of the following argument to show that every $n \times n$ matrix A with entries in $\mathbb{H}$ has an eigenvalue in $\mathbb{H}$. (The usual argument over $\mathbb{C}$ involving roots of the characteristic polynomial does not work due to the lack of a good quaternionic determinant function.) For $t \in [0, 1]$ and $\lambda \in S^3 \subset \mathbb{H}$, consider the matrix $t\lambda I + (1-t)A$. If A has no eigenvalues, this is invertible for all t . Thus the map $S^3 \rightarrow GL_n(\mathbb{H})$, $\lambda \mapsto \lambda I$, is nullhomotopic. But by the preceding problem and Exercise 10(b) in §3.C, this map represents n times a generator of $\pi_3 GL_n(\mathbb{H})$.